

THARUN KUMAR KSHEERASAGAR

Senior Architect

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SUMMARY

System Architecture and Performance Evaluation Engineer with over a decade driving pre-silicon validation and compute acceleration at NVIDIA, AMD, and Samsung. Expertise in hardware-software co-design, hybrid simulation, and architecting LPUs, NPUs, and GPUs to eliminate system bottlenecks. Recent work spans advanced AI infrastructure, from integrating NVIDIA iGPUs into MediaTek SoCs to modeling next-gen LPU- and NPU-based inference accelerators.

WORK EXPERIENCE

Stealth AI Hardware Startup (Operations Ceased)

Hyderabad, India

Senior Architect

Apr 2026 – Sept 2026

- Led a two-engineer team while developing a configurable SystemC performance model delivering exact cycle-by-cycle execution for a Groq-inspired AI inference accelerator card tailored for Transformer and LLM workloads.
- Provided a golden reference model for the compiler team, establishing predictable, low-latency execution frameworks for real-time Transformer workloads.
- Prototyped an intuitive web-based data-flow visualization tool using agentic AI to track real-time packet movement across hardware blocks, accelerating compiler debug turnaround times.

NVIDIA

Hyderabad, India

Senior Architect

Jan 2025 – Apr 2026

- Led system-level validation and cross-functional debug for NVIDIA's next-generation N2X AI platform.
- Spearheaded the complex integration of an NVIDIA iGPU into a MediaTek SoC, driving virtual platform bring-up to achieve successful system Linux and GPU firmware boot in an accelerated 4-month timeframe.

AMD

Hyderabad, India

Member of Technical Staff Software Engineer

June 2019 – Jan 2025

- Designed and delivered a custom multi-threaded simulation kernel within a strict six-month timeframe, maintaining SystemC semantics while accelerating simulation speeds by up to 10x.
- Drove the integration of AMD's AI Engine (AIE1/AIE2) Neural Processing Unit (NPU) models into the new infrastructure, exceeding the initial proof-of-concept scope by transpiling and fine-tuning a cycle-accurate model into a functional, untimed simulation model.
- Developed a comprehensive software debug framework tailored for AI workloads, providing robust tooling to troubleshoot and optimize Convolutional Neural Network (CNN) architectures such as ResNet and YOLO.
- Developed a custom memory model in Xilinx Vitis for advanced performance analysis of Alveo data center accelerator cards, supporting high-performance compute workloads across Machine Learning (ML) inference and FinTech algorithmic trading.
- Accelerated Xilinx Vitis hardware emulation speed by 6x (doubling the MVP target of 3x), significantly reducing simulation and debug turnaround times for complex architectures.
- Engineered a scalable, latency-accurate I/O emulation feature for the [Xilinx Accelerated Algorithmic Trading \(AAT\)](#) framework, enabling cycle-accurate simulations using PCAP inputs for precise packet-level visibility.

SAMSUNG

Bengaluru, India

Associate Staff Engineer

June 2016 – June 2019

- Architected a Cycle-Accurate Hybrid Simulation Framework that synchronized SystemC and FPGA environments, enabling seamless emulation of complex, distributed designs.
- Developed critical cycle-accurate models and transactors, including an AXI3 Interconnect SystemC model and AXI-to-TLM transactors, to drive advanced performance measurement for memory subsystems.
- Engineered a custom Co-Simulation framework leveraging Interconnect WorkBench (IWB/Incisive), SystemC, and NCSim to streamline cross-platform verification and design testing.

PATENTS

- **Model Level Debugging of Machine Learning Designs on Neural Processing Units** (Published, [US-20260186951-A1](#))
- **Multi-Threaded Cycle Accurate Architecture Simulation** (Published, [US20240184616-A1](#))
- **Data traffic injection for simulation of circuit designs** (Issued, [US-11630935-B1](#))
- **Circuit design simulation and clock event reduction** (Issued, [US-12086521-B2](#))
- **Adaptive Simulation Framework for Switching Between Functional and Cycle-accurate Execution in Real-Time** (Pending)

PUBLICATIONS

- **Performance analysis of DS-CDMA using different chaotic sequences** – [IEEE](#), 2016 Int. Conf. on Wireless Communications, Signal Processing and Networking (WiSPNET), Mar 2016, pp. 2421–2425

CERTIFICATIONS

- **GPU Architectures and Programming** – NPTEL (Topper) *May 2022*
- **Multi-Core Computer Architecture – Storage and Interconnects** – NPTEL (Elite Gold) *Oct 2022*
- **Operating System Fundamentals** – NPTEL (Topper) *Nov 2021*
- **Modern C++ Concurrency in Depth** – Udemy *Oct 2020*

SKILLS & AWARDS

- **Languages & Libraries:** C++, SystemC, Python
- **Awards:** AMD Executive Spotlight Award, Xilinx Quarterly Award, Samsung Employee of the Month
- **Tools & Frameworks:** Synopsys Virtualizer Studio, AMD Vivado / Vitis, AMD xsim, Cadence IWB / Incisive, NCSim

EDUCATION

M.Tech in Software Systems CGPA: 8.19	BITS Pilani (WILP)
B.Tech in Electronics & Communication Engineering CGPA: 8.94 <i>Consistent Institute Merit Scholar</i>	NIT Warangal